



Loading data

```
import pandas as pd
from google.colab import files
csv = files.upload()
iris_df = pd.read_csv("Iris.csv")
print(iris_df)
```

No files selected.

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	\
0	1	5.1	3.5	1.4	0.2	
1	2	4.9	3.0	1.4	0.2	
2	3	4.7	3.2	1.3	0.2	
3	4	4.6	3.1	1.5	0.2	
4	5	5.0	3.6	1.4	0.2	
..	...	...	...	...	...	
145	146	6.7	3.0	5.2	2.3	
146	147	6.3	2.5	5.0	1.9	
147	148	6.5	3.0	5.2	2.0	
148	149	6.2	3.4	5.4	2.3	
149	150	5.9	3.0	5.1	1.8	

	Species
0	Iris-setosa
1	Iris-setosa
2	Iris-setosa
3	Iris-setosa
4	Iris-setosa
..	...
145	Iris-virginica
146	Iris-virginica
147	Iris-virginica
148	Iris-virginica
149	Iris-virginica

[150 rows x 6 columns]

TASK 1: Import 'iris' data (.csv format) Display

1. Display the last 10 elements
2. Copy the second column(width) to a list
3. Display the first 10 elements
4. Create a new column called Species_label
5. change name in Species to 0,1,2 in Species_label

```
print(iris_df.iloc[-10:])
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	\
140	141	6.7	3.1	5.6	2.4	
141	142	6.9	3.1	5.1	2.3	
142	143	5.8	2.7	5.1	1.9	
143	144	6.8	3.2	5.9	2.3	
144	145	6.7	3.3	5.7	2.5	
145	146	6.7	3.0	5.2	2.3	
146	147	6.3	2.5	5.0	1.9	
147	148	6.5	3.0	5.2	2.0	
148	149	6.2	3.4	5.4	2.3	
149	150	5.9	3.0	5.1	1.8	

	Species
140	Iris-virginica
141	Iris-virginica
142	Iris-virginica
143	Iris-virginica
144	Iris-virginica
145	Iris-virginica
146	Iris-virginica
147	Iris-virginica
148	Iris-virginica
149	Iris-virginica

```
# need to convert to list because normally pd[<column name>] return pd
col2 = list(iris_df['SepalWidthCm'])
print(col2)
```

```
[3.5, 3.0, 3.2, 3.1, 3.6, 3.9, 3.4, 3.4, 2.9, 3.1, 3.7, 3.4, 3.0, 3.0,
```

```
# Display the first 10 rows
iris_df.iloc[:10]
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
5	6	5.4	3.6	1.7	0.4	Iris-setosa

5	6	5.4	3.9	1.7	0.4	setosa
6	7	4.6	3.4	1.4	0.3	Iris-

```
# Copy the Species to new 'Species_label' column
iris_df['Species_label'] = iris_df['Species']
print(iris_df)
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	\
0	1	5.1	3.5	1.4	0.2	
1	2	4.9	3.0	1.4	0.2	
2	3	4.7	3.2	1.3	0.2	
3	4	4.6	3.1	1.5	0.2	
4	5	5.0	3.6	1.4	0.2	
..	...	...	...	...	...	
145	146	6.7	3.0	5.2	2.3	
146	147	6.3	2.5	5.0	1.9	
147	148	6.5	3.0	5.2	2.0	
148	149	6.2	3.4	5.4	2.3	
149	150	5.9	3.0	5.1	1.8	

	Species	Species_label
0	Iris-setosa	Iris-setosa
1	Iris-setosa	Iris-setosa
2	Iris-setosa	Iris-setosa
3	Iris-setosa	Iris-setosa
4	Iris-setosa	Iris-setosa
..	...	...
145	Iris-virginica	Iris-virginica
146	Iris-virginica	Iris-virginica
147	Iris-virginica	Iris-virginica
148	Iris-virginica	Iris-virginica
149	Iris-virginica	Iris-virginica

[150 rows x 7 columns]

```
Species = iris_df['Species_label'].unique() # Unique() return numpy array
for i in range(len(Species)):
    iris_df['Species_label'] = iris_df['Species_label'].replace(to_replace=Species[i], value=i)
display(iris_df['Species_label'])
```

```
/tmp/ipython-input-3190910974.py:3: FutureWarning: Downcasting behavior
iris_df['Species_label'] = iris_df['Species_label'].replace(to_replace=Species[i], value=i)
```

	Species_label
0	0
1	0
2	0
3	0
4	0

```

      ...
145      2
146      2
147      2
148      2
149      2

```

150 rows × 1 columns

dtype: int64

TASK2 image.png

1. Find mean/median/min/max/average for each column
2. Print mean,var,min,max group by Species label
3. Plot (scatter) each data based on each type

```

display(iris_df.mean(numeric_only=True))
display(iris_df.median(numeric_only=True))
display(iris_df.min(numeric_only=True))
display(iris_df.max(numeric_only=True))
#display(iris_df.avg(numeric_only=True))

```

	0
Id	75.500000
SepalLengthCm	5.843333
SepalWidthCm	3.054000
PetalLengthCm	3.758667
PetalWidthCm	1.198667
Species_label	1.000000

dtype: float64

	0
Id	75.50
SepalLengthCm	5.80
SepalWidthCm	3.00

```

PetalLengthCm    4.35
PetalWidthCm    1.30
Species_label    1.00

```

dtype: float64

```

      0
-----
Id    1.0

```

```

SepalLengthCm    4.3
SepalWidthCm    2.0
PetalLengthCm    1.0
PetalWidthCm    0.1
Species_label    0.0

```

dtype: float64

```

      0
-----
Id    150.0
SepalLengthCm    7.9
SepalWidthCm    4.4
PetalLengthCm    6.9
PetalWidthCm    2.5
Species_label    2.0

```

dtype: float64

```

def print_format(df,stat):
    print(f"{'-'*40} {stat} {'-'*40}\n") # fixed number of dashes
    func = getattr(df.groupby('Species'), stat)
    display(func(numeric_only=True))
    print('\n')
iris_df.groupby('Species').mean(numeric_only=True)
print_format(iris_df, 'mean')
print_format(iris_df, 'var')
print_format(iris_df, 'min')
print_format(iris_df, 'max')
#print(iris_df.groupby("Species").mean())

```

```

----- mean -----
Td SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm

```

Species	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Iris-setosa	25.5	5.006	3.418	1.464	0.24
Iris-versicolor	75.5	5.936	2.770	4.260	1.32
Iris-virginica	125.5	6.588	2.974	5.552	2.02

----- var -----					
Species	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Iris-setosa	212.5	0.124249	0.145180	0.030106	0.01149
Iris-versicolor	212.5	0.266433	0.098469	0.220816	0.03910
Iris-virginica	212.5	0.404343	0.104004	0.304588	0.07543

----- min -----					
Species	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Iris-setosa	1	4.3	2.3	1.0	0.1
Iris-versicolor	51	4.9	2.0	3.0	1.0
Iris-virginica	101	4.9	2.2	4.5	1.4

----- max -----					
Species	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm

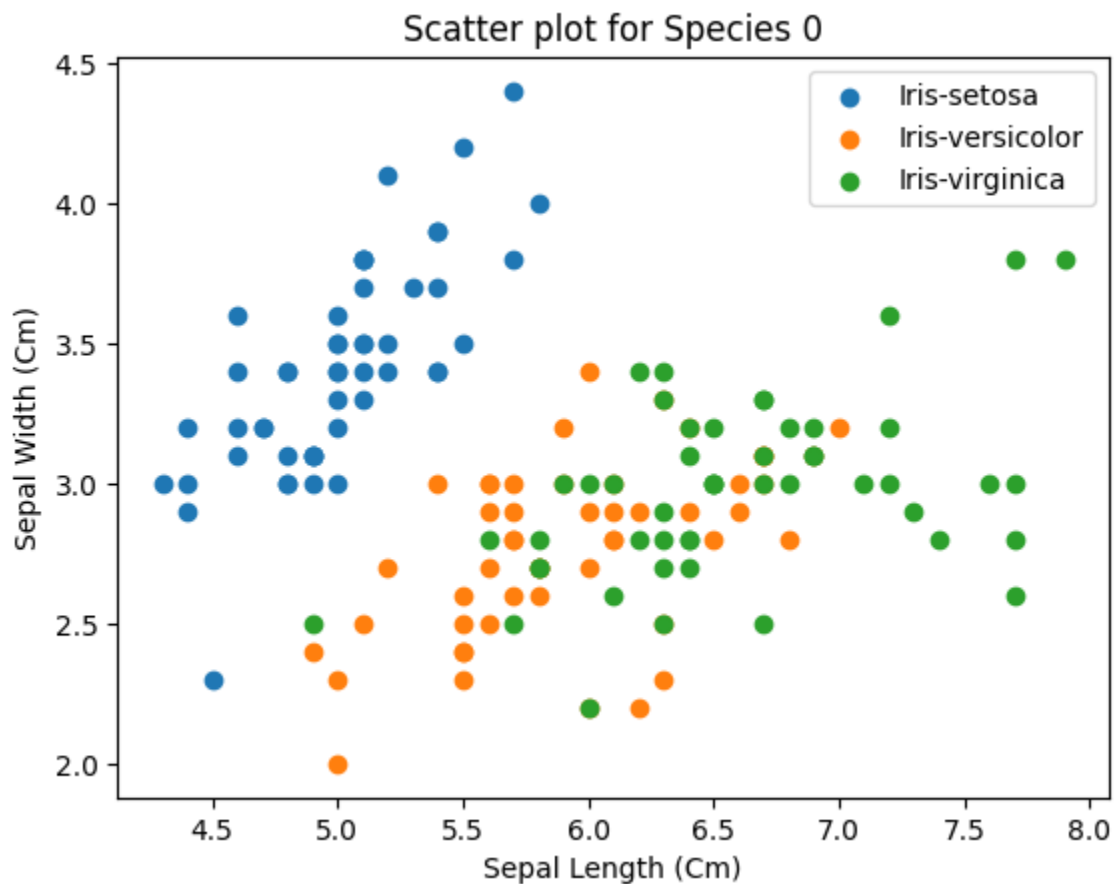
```
#Visualization plotting scatter plot
import matplotlib.pyplot as plt
df0 = iris_df[iris_df['Species_label']==0]
x = df0['SepalLengthCm']
```

```

x = df0['SepalLengthCm']
y = df0['SepalWidthCm']
plt.scatter(x, y, label=df0['Species'].unique()[0])
df1 = iris_df[iris_df['Species_label']==1]
x = df1['SepalLengthCm']
y = df1['SepalWidthCm']
plt.scatter(x, y, label=df1['Species'].unique()[0])
df2 = iris_df[iris_df['Species_label']==2]
x = df2['SepalLengthCm']
y = df2['SepalWidthCm']
plt.scatter(x, y, label=df2['Species'].unique()[0])

plt.xlabel('Sepal Length (Cm)')
plt.ylabel('Sepal Width (Cm)')
plt.title('Scatter plot for Species 0')
plt.legend()
plt.show()

```



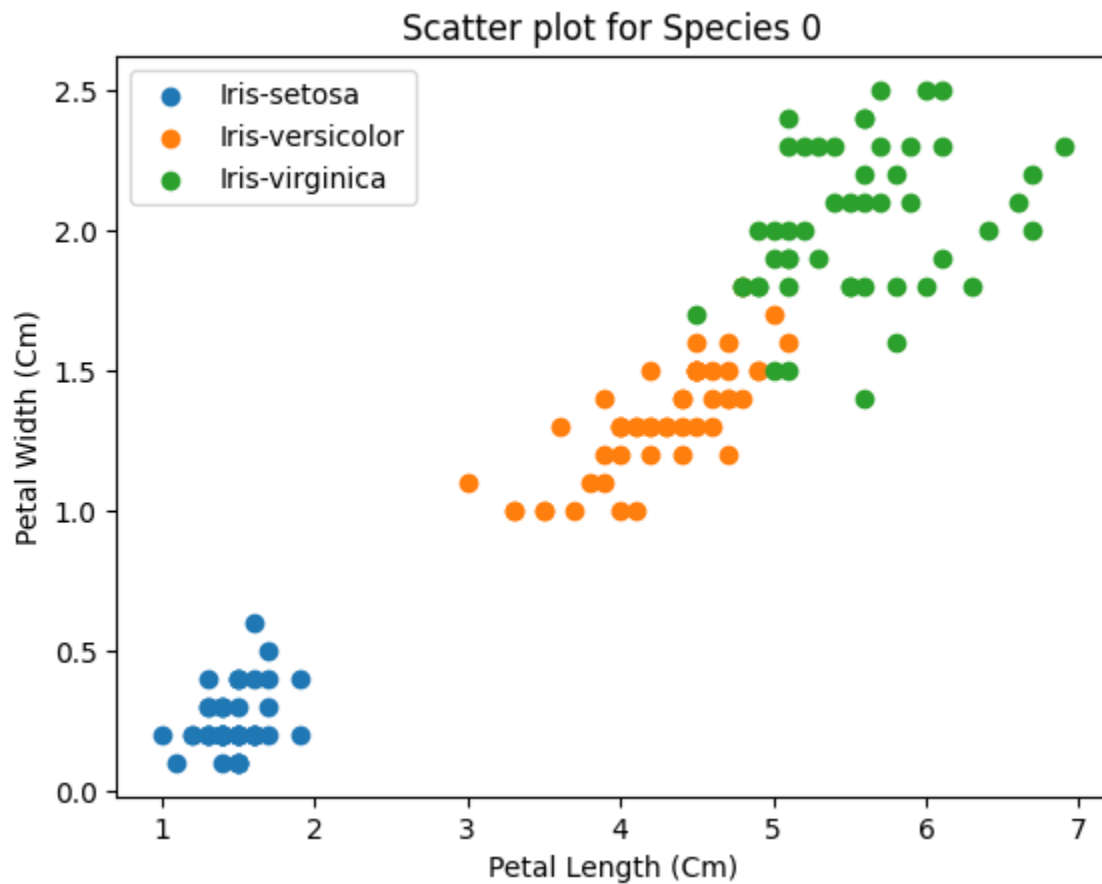
```

df0 = iris_df[iris_df['Species_label']==0]
x = df0['PetalLengthCm']
y = df0['PetalWidthCm']
plt.scatter(x, y, label=df0['Species'].unique()[0])
df1 = iris_df[iris_df['Species_label']==1]
x = df1['PetalLengthCm']
y = df1['PetalWidthCm']
plt.scatter(x, y, label=df1['Species'].unique()[0])

```

```
plt.scatter(x, y, label=df1['Species'].unique()[0])
df2 = iris_df[iris_df['Species_label']==2]
x = df2['PetalLengthCm']
y = df2['PetalWidthCm']
plt.scatter(x, y, label=df2['Species'].unique()[0])

plt.xlabel('Petal Length (Cm)')
plt.ylabel('Petal Width (Cm)')
plt.title('Scatter plot for Species 0')
plt.legend()
plt.show()
```



TASK3

Try using describe() function

```
iris_df.describe()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763780

min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

TASK4

Show 1 animal each from

```
# Sorry I don't understand the question
```